

VAIBHAV VIKAS RANJAN

+91 7080802844

vaivikran@gmail.com

LinkedIn

GitHub

Portfolio

Education

International Institute of Information Technology, IIIT-NR

Nov 2022 - July 2026

B.Tech Data Science and Artificial Intelligence

Naya Raipur, Chhattisgarh

Experience

Accure Inc.

Jan 2026 – July 2026

AI Engineer Intern

Bangalore, India

- Improved **LLM reasoning reliability** for an enterprise **generative AI** platform (AccureIQx) using a structured multi-criteria **evaluation and validation** process, by designing and governing **agentic AI workflows** and **prompt-engineering** pipelines for **enterprise search** and reasoning use cases.
- Strengthened adversarial robustness of **agentic reasoning** systems, improving **anomaly-detection** performance across **five stress-test scenarios**, by engineering synthetic cybersecurity incident datasets with realistic log data.
- Cut manual weekly reporting time by **35%**, saving analyst hours, by building automated **Python (Pandas, NumPy)** analytics pipelines integrating multi-source EDA data via **REST APIs**.

Taskar – India's 1st Healthcare Mall

May 2025 – July 2025

Software Developer Intern

Remote, India

- Reduced manual reporting workflows by **40%**, cutting process time, by building **microservice-based RESTful API** integrations to retrieve and analyze live **healthcare and inventory data** for business stakeholders.
- Improved **analytics dashboard** load times by **30%**, per page-render benchmarks, by optimizing **UI rendering** and **asynchronous data fetching** with **React Hooks**.
- Increased user engagement by **15%**, confirmed by **A/B test** results, by redesigning dashboard layouts based on internal stakeholder and cross-team feedback.

Research Designs and Standards Organisation (R.D.S.O)

May 2024 – July 2024

Software Engineering Intern

Lucknow, India

- Supported **500+ concurrent users** at **99.9% uptime**, verified through system load testing, by engineering a responsive, **service-oriented attendance tracking system (HTML, CSS, JavaScript, PHP, MySQL)**.
- Cut login times by **60%** for **1,000+ daily users**, based on authentication response benchmarks, by designing an optimized **authentication workflow** with **AJAX-based** real-time interactivity.
- Boosted user retention **20%** and validated production-grade performance under high concurrency via **Apache JMeter** load and stress testing across peak-usage scenarios.

Projects

CFR Compliance Checker: Agentic Contract Auditor | *Python, FastMCP, Agno, Ollama, Pydantic* | [Github](#)

- Built a custom **MCP server** wrapping the official **eCFR REST API (7 typed MCP tools** for regulatory search and retrieval) and architected an **end-to-end agentic pipeline** using an **Agno-based LLM compliance agent** on local **Ollama (Llama 3.1)**, producing validated, structured compliance results per clause.
- Optimized the pipeline with **concurrent CFR retrieval** and evaluation, reducing end-to-end runtime by **65%** (12 min → 4:10) on a **24-clause** government contract, with async I/O and caching for low-latency responses.

Incident Response Engine | *Python, Agno, FastMCP, Kafka, ChromaDB, FastAPI, Pydantic* | [Github](#)

- Built a **multi-agent incident-response** pipeline with a runtime-validated **Kafka** layer, using a **z-score anomaly gate** and two-phase Investigator Agent separating **MCP tools** from LLM analysis.
- Implemented a **human-in-the-loop safety gate** via Slack + FastAPI/React, requiring **ApprovalDecision** sign-off, with sandboxed MCP diagnostics and a **147-test pytest suite** validating security, contracts, orchestration.

Multi-Agent Orchestrator: GitHub Issue → PR Pipeline | *Python, LangGraph, Docker, GitHub API* | [Github](#)

- Architected a **multi-agent LangGraph pipeline** (Researcher, Planner, Coder, Verifier, PRWriter) converting **GitHub Issues** into verified **Pull Requests**, using conditional **retry/escalation routing** grounded in real repository context.
- Enforced a **human-in-the-loop approval gate** and **Docker-sandboxed verification** before code execution, with idempotent **GitHub PR publication** and secret-redacting logs for production reliability.

Technical Skills and Interests

Languages: Python, SQL, C++, JavaScript, HTML/CSS

Technologies/Frameworks: NumPy, Pandas, Matplotlib, PyTorch, TensorFlow, Scikit-learn, OpenCV, LangChain, LangGraph, Streamlit, FastAPI, MCP, Chainlit, Agno, PostgreSQL

Tools & Platforms: Docker, AWS, Git, Tableau, Hugging Face, ChromaDB

AI/ML & familiar Domain: Generative AI, LLMs, RAG (Retrieval-Augmented Generation), NLP, Machine Learning, Computer Vision, Anomaly Detection, Deep Learning, Full Stack Development, Data Engineering, Prompt Engineering, Data Analysis